

Session 6 — The Moon arrives

Practical Astronomy Intensive

Name _____

Date _____

6 questions 15 minutes

Terminator over full moon, impacts over volcanoes, 59% over 50% — plus the two things they should be able to DO: drive the scope to the terminator and bring back a focused phone frame of it.

*Circle one letter for each multiple-choice question. Write short answers on the lines. Tasks marked **Practical** are done, not written — your assessor will watch.*

- 1 A visitor asks when to bring their new telescope for the best view of lunar craters. What do you tell them?
 - a At full moon — the whole disc is lit, so you see the most
 - b Any phase, but aimed at the bright middle of the disc where the light is strongest
 - c Around first quarter or a crescent, and look along the terminator, where low sunlight throws long shadows off every crater rim
 - d At full moon, with a Moon filter fitted; the filter is what brings the craters out
- 2 Mare Tranquillitatis is a large dark patch, smoother than the bright surrounding highlands. What is it?
 - a A young surface, smooth because lunar weather has worn the craters away
 - b A sea, or the dry bed of one that has since evaporated
 - c An ancient basalt lava plain flooding an old impact basin — dark and smooth because the lava resurfaced it
 - d A field of collapsed volcanic calderas, which is also what the craters are
- 3 The Moon is tidally locked, keeping one face towards Earth. Over many months, how much of its surface can we actually see from Earth, and why?
 - a About 59% — libration rocks the Moon side to side and nods it up and down, letting us peek round each limb in turn
 - b Exactly 50% — tidal locking means the same half faces us and the rest never does
 - c 50%, since the other 41% is the permanently dark side that never receives sunlight
 - d It varies unpredictably, because the Moon wobbles randomly on its axis

- 4 A student has a good phone and tonight's crescent Moon in the eyepiece. Which is the honest account of what that phone can and cannot do?
- a Zoom the phone in far enough and it will pull M31 out of the sky the same way it gets the Moon
 - b Held at the eyepiece on a bracket it will give a genuinely good Moon shot on the first try; with night mode on a tripod it records constellations and, from a dark site, the Milky Way — but it will not reach a galaxy without a telescope in front of it
 - c A phone cannot do real astrophotography; it is a toy until you buy a DSLR
 - d It will work at the eyepiece if you hold it steady enough; a bracket is for people with shaky hands

- 5 The Moon is up and the scope is yours. Find the terminator, put a crater group on it in the eyepiece, and sketch what you see in five minutes — then tell the instructor why you chose that spot on the disc.

Practical — performed for your assessor.

- 6 Fit the phone adapter to the eyepiece and bring back one focused shot of the crater terminator — sharp enough that individual crater shadows are visible.

Practical — performed for your assessor.